

A Systematic Review of Machine Learning Approaches for Crop Recommendation Systems

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Abstract—Machine Learning based crop recommendation systems have gathered a good amount of attention in precision agriculture due to their ability to improve crop yield and productivity, along with optimized resource utilization and reducing financial risks for farmers. Such systems make use of various soil characteristics and climatic parameters to recommend which crop would be ideal to grow in given set of conditions. This paper presents a review of recent research works and studies that are focused on crop recommendation using machine learning. We examined and reviewed ten such recent works with respect to the learning models, performance techniques and data acquisition techniques. After reviewing these studies, it has been observed that ensemble and boosting-based algorithms like Random Forest, XGBoost and CatBoost have consistently achieved higher performance and accuracy than other conventional learning methods. Along with the performance of learning models, the review highlights some key limitations in many existing systems, including collecting real-time data and the use of static datasets. Based on these observations, the paper focuses on the need for automated data collection and real-time environment integration to bridge the gap between prototypes and practical deployment. Overall, this review aims to summarize ongoing trends, identify limitations and to outline future research directions for developing a reliable crop recommendation system.

Keywords— Crop recommendation, machine learning, IoT-based agriculture, ensemble learning, real-time data integration.

I. INTRODUCTION

From its very inception, agriculture has been instrumental in providing food security and ensuring economic stability for both farmers and the nation. Over the last few years, farmers have faced many difficulties as a result of changes in climatic conditions and soil quality, resulting in poor crop selection decisions. Traditional approaches to decision-making, which are based on experience or previous trends, tend to become inadequate when dealing with the complexity of the situation in modern agriculture. Consequently, in recent years, there has been a greater interest in data-driven approaches that can assist farmers in making better decisions.

The development of machine learning-based crop recommendation systems is one of the solutions in this field. Such systems allow analyzing the composition of soil nutrients, including nitrogen, phosphorus, and potassium, as well as climatic indicators like temperature, precipitation, humidity, and soil acidity to recommend crops that can be grown successfully in a particular region. The main purpose of using such systems is to minimize the possibility of crop failure, thereby increasing yields and promoting sustainable agricultural practice. Despite numerous studies reporting high accuracy rates of predictions, certain issues still arise.

This paper aims at reviewing and comparing recent studies devoted to the application of machine learning-based crop recommendation systems. In particular, it highlights various learning approaches employed, performance trends, and limitations observed, with an emphasis on the existing research gap and future research directions.

II. LITERATURE SURVEY

In recent times, a significant amount of research has been done and focused on applying machine learning techniques to crop recommendation systems so as to support precision agriculture. While reading and reviewing these studies we observed that most of them use soil nutrients and climatic parameters to predict suitable crops. This section reviews key contributions of various researchers, focusing on their methodologies, performance and limitations.

Pathan et al. [1] proposed a machine learning-based crop recommendation system focusing at assisting Indian farmers in crop selection. The authors simply assessed many supervised machine learning models like Decision Tree, Support Vector Machine, Logistic Regression and Gaussian Naive Bayes. Their experimental results reported high predictive accuracy, with decision tree and Gaussian Naive Bayes achieving best performance (99.3%). They used soil nutrients like NPK along with temperature, pH, humidity and rainfall. We observed that it heavily relies on a small static dataset and does not incorporate real time field and environmental data.

Apat et al. [2] presented an AI-based crop recommendation framework that evaluated multiple machine learning and boosting algorithms by using the same soil and climatic features. To enhance the robustness of the model, the authors applied SMOTE and feature selection techniques. Their results specified that ensemble and boosting models outperformed typical classifiers, with CatBoost achieving the highest accuracy of around 99.5%. However, the study is limited because of its lack of real-time integration, dependence on historical data, and high computational complexity due to the use of advanced boosting models.

Kale and Mohapatra [3] developed a crop recommendation system using Random Forest, KNN, SVM, and Artificial Neural Networks. On the basis of their experiments, Random Forest achieved the highest accuracy (99.22%) with lowest training time. The authors also showed a web-based implementation of user interface. Despite all these strengths, we noted that this study relies on historical data and cannot be implemented in region-specific and real-time data.

Afzal et al. [4] proposed a hybrid ensemble model that combines Random Forest and XGBoost to improve accuracy of crop recommendation systems. They used a dataset of 2,200 rows and this hybrid model proposed accuracy of 98%, thereby outperforming other individual models. Although this method improves prediction performance, the authors did acknowledge that such complex learning techniques increase computational cost and require careful tuning, which may limit the scalability.

Gosai et al. [5] presented a combination of machine learning with plant disease detection as an intelligent crop recommendation system. The authors evaluated a wide range of classifiers and reported that ensemble based models achieved the best results, with XGBoost reaching an accuracy of 99.31%. Along with this, the authors also highlighted the future scope of integrating sensor-based data as having significant potential. However, their work lacks real-world field validation and cannot be generalized across different regions and seasons.

Gupta et al. [6] focused on integration of IoT technology with machine learning to support real - time crop prediction. They tested several classifiers and used IoT sensors to collect real-time temperature and humidity data. The Decision tree classifier had the best accuracy in those tests. We observed that their system uses only a little amount of sensing infrastructure and still relies on historical data, even though adding sensor data can make it more useful in real life usage.

Dey et al. [7] performed a comparative analysis of various machine learning models for agriculture and horticultural crops. Instead of using one generic model, they built specialized ones per crop for better predictions. The authors said that XGBoost was best amongst all the performance models, but more environmental data and better deployment methods are needed for it to be used in real life.

El Barrak et al. [8] tested seven different supervised learning algorithms and found that Random Forest, Naive Bayes, and LightGBM always did better than the others. Support Vector Machine performed very poorly, which shows that not all standard machine learning algorithms work well with agricultural datasets without necessary tuning.

Islam et al. [9] demonstrated a complete IoT-based system that combines machine learning with real-time soil monitoring to recommend crops and fertilizers. The system used sensor-based data along with cloud communication to get accurate results. It also added a mobile application to interact with users. The authors identified frequent misclassifications in their work and suggested that future work should include satellite images and more general weather data.

Finally, Elbasi et al. [10] performed a big study by comparing 15 machine learning algorithms and found that Bayes Net and Naive Bayes were the most accurate algorithms. The study also showed that expensive soil testing might not always be needed and smaller sets of features could still give good results. But the authors also pointed out problems with data quality, sensor cost, and deploying the system on a large scale.

Overall, it was found that all the models used in the ensemble were always accurate, and the use of data collection through IoT technology ensured the results to be relevant. However, most of the previous works relied on the use of static data, failed to test their findings in the field for an extended period of time, and faced difficulties in implementation.

III. COMPARATIVE STUDY AND ANALYSIS

Upon analyzing all these selected research works dealing with crop recommendation using machine learning algorithms, we noticed that there is a great similarity in terms of performance capabilities, data usage, and applicability of such systems in reality. It should be noted that this section does not focus on each research work individually but rather concentrates on comparing the approaches used and demonstrates how one research work's shortcomings are addressed by another.

A. Comparison of Learning Models:

The models such as Decision Tree, Nave Bayes, Support Vector Machine, and Logistic Regression, which are considered classical supervised learning models, have been widely adopted as benchmarks for classification tasks [1], [3], [5]. These models produced satisfactory performance when applied to datasets that contained structured data related to soil and climate features. Decision Tree-based models are particularly effective due to their ability to handle nonlinear dependencies.

Nevertheless, having analyzed many studies, we have noticed that ensemble-based approaches always outperform traditional classifiers. In particular, many researchers [3], [4], [7], and [8] note that Random Forest can be considered as a robust and trustworthy classifier. This algorithm is suitable for agricultural applications because it minimizes the risk of overfitting and maintains low computational complexity. Meanwhile, there are some studies [2], [7] that prove boosting algorithms such as XGBoost and CatBoost to be slightly more accurate than others; however, they require more resources and tuning.

B. Characteristics and Acquisition of Data:

Other significant observations drawn from the literature review include the fact that most of the existing crop recommendation systems rely on static or historic data [1], [2], [3], [7], and [10]. This type of data is ideal for testing purposes and accuracy, but it does not depict the dynamic nature of soil and environment.

IoT-based sensor data studies [6], [9] that focus on real-time monitoring overcome this challenge. The application of such methodologies reduces the necessity of manual soil and weather testing, making it more practical. However, they pose challenges related to the cost of sensors, infrastructure issues, and maintenance of the system, which need to be considered.

C. Generalization of Models and Dependency on Features:

Another aspect that we observed in these papers is the use of one model trained on different crop data sets. This will lead to confusion in classifying the crops that have similar requirements [1], [5]. One way of addressing this issue is through the use of category-based models that increase accuracy but increase complexity as well [7]. Moreover, most of the systems depend mainly on NPK and weather parameters, resulting in increased accuracy but also making soil testing centers more significant [1]-[5]. The reduced feature approach somewhat addresses this drawback but can have lesser efficiency in the field [10].

D. Key Comparative Observations:

From the comparison, it is clear that ensemble modeling algorithms like Random Forest offer a perfect balance between accuracy and computation, whereas IoT-enabled data collection makes use of dynamic collection rather than the static data sets used previously.

The above-mentioned considerations indicate that future crop recommendation systems must be designed in such a way that learning models along with data acquisition systems are incorporated in them.

TABLE I. COMPARATIVE ANALYSIS OF MACHINE LEARNING MODELS FOR CROP RECOMMENDATION SYSTEMS

Ref.	ML Techniques Used	Best Performing Model / Accuracy	Key Findings & Limitations
[1]	DT, SVM, LR, GNB	DT / GNB – 99.3%	Classical ML achieves high accuracy; static dataset, no real-time data or field validation.
[2]	LR, DT, GNB, RF, CatBoost	CatBoost – ~99.5%	Boosting models outperform others; high computation cost, no real-time or regional validation.
[3]	RF, KNN, SVM, ANN	RF – 99.22%	RF offers high accuracy and low training time; lacks real-time data and advanced hybrid models.
[4]	RF, XGB, LR, MLP, Extra Trees	RFXG – 98%	Hybrid ensemble improves performance; computationally expensive, limited dataset evaluation.
[5]	DT, NB, SVM, LR, RF, XGB, CNN	XGBoost – 99.31%	Ensemble models effective and disease detection integrated; lacks field validation and economic factors.
[6]	DT, KNN, SVM	DT – Highest	IoT-based real-time sensing improves reliability; limited sensors, no adaptive/deep learning.
[7]	DT, RF, KNN, SVM, XGB	XGBoost – ~99.3%	Separate crop-category models outperform mixed models; limited scalability and deployment.
[8]	DT, RF, NB, KNN, LR, SVM, LightGBM	RF / NB / LightGBM – High	Tree-based and Bayesian models effective; SVM performs poorly without tuning.
[9]	CatBoost, RF	CatBoost / RF – ~99%	Real-time IoT-based crop and fertilizer recommendation; minor misclassification, limited weather data.
[10]	NB, RF, MLP, Hoeffding Tree, Bayes Net	Bayes Net – 99.59%	Feature reduction retains high accuracy; relies on static data, limited real-world validation.

IV. CONCLUSION

A total of ten articles were reviewed on the basis of crop recommendation systems using machine learning techniques to understand how models work and solve problems through adopting different techniques. After reviewing the articles, we came to know that machine learning techniques are highly efficient in selecting the crops to be grown. Algorithms like random forest and XGBoost have been found to be efficient models. The machine learning models are efficient enough to analyze the parameters like soil nutrients, soil pH, temperature, and rainfall better than the traditional approaches.

However, one limitation that is often observed in most research papers is their high reliance on data sets that are either static, historical, or manually collected. Although this approach

serves its purpose in laboratory settings, it fails to account for the dynamic nature of actual agricultural environments. This has greatly limited the applicability of many systems suggested by researchers.

In general, the findings from this study indicate that there is no need for future studies on crop recommendation systems to concentrate only on increasing the accuracy of machine learning models. There should be more emphasis on making the process practical by acquiring dependable data and creating effective systems. The combination of machine learning models with an Internet of Things-based sensor system for automated and real-time acquisition of environmental data is a promising approach towards bridging the existing gap.

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